The Chain Rule & Directional Derivatives

ANSWER KEY

Section 1: Multivariable Chain Rule

1 If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ ✓

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

By the multivariable chain rule, rates of change sum across all intermediate paths: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.

2 Given $z = x^2y$, $x = t^3$, and $y = t^2$, find $\frac{dz}{dt}$ at $t = 1$.

a. 8 ✓

b. 6

c. 5

d. 12

$z = (t^3)^2(t^2) = t^6 \cdot t^2 = t^8$. Therefore $\frac{dz}{dt} = 8t^7$. At $t = 1$, $\frac{dz}{dt} = 8(1)^7 = 8$.

3 If $z = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta$, what is $\frac{\partial z}{\partial r}$?

a. $\frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$ ✓

b. $-r\frac{\partial z}{\partial x}\sin\theta + r\frac{\partial z}{\partial y}\cos\theta$

c. $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\cos\theta + \sin\theta$

$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$.

Section 2: The Gradient Vector

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4 What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$ ✓

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

The gradient of a multivariable function is the vector consisting of its first-order partial derivatives: $\nabla f = [f_x, f_y]$.

5 Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$ ✓

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

$f_x = 6xy \Rightarrow f_x(2, -1) = 6(2)(-1) = -12$. $f_y = 3x^2 + 4y \Rightarrow f_y(2, -1) = 3(4) + 4(-1) = 12 - 4 = 8$. Gradient is $[-12, 8]$.

6 What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ **is perpendicular (orthogonal) to the tangent line of the level curve** ✓

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

The gradient vector at any point is always perpendicular to the level curve (or level surface) passing through that point.

Section 3: Directional Derivatives

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7 What is the formula for the directional derivative $D_{\mathbf{u}}f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ ✓

b. $D_{\mathbf{u}}f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}}f = \|\nabla f\|\|\mathbf{u}\|$

d. $D_{\mathbf{u}}f = \frac{\nabla f \cdot \mathbf{u}}{\|\nabla f\|}$

The rate of change of f in the direction of unit vector $\mathbf{u}$ is the dot product of the gradient with the unit vector: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

8 If $\nabla f(1, 2) = [4, -3]$ and $\mathbf{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$, find $D_{\mathbf{u}}f(1, 2)$.

a. 0 ✓

b. 5

c. $\frac{24}{5}$

d. -1

$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = (4)\left(\frac{3}{5}\right) + (-3)\left(\frac{4}{5}\right) = \frac{12}{5} - \frac{12}{5} = 0.$

9 Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$?

a. **To ensure the derivative measures rate of change per unit of geometric distance** ✓

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

Normalizing $\mathbf{u}$ ensures that the directional derivative accurately reflects the instantaneous rate of change per unit distance along that direction.

Section 4: Direction of Maximum Increase & Steepest Ascent

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10 In which direction does a differentiable function f increase most rapidly at a point?

a. **In the direction of the gradient vector ∇f** ✓

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

Since $D_{\mathbf{u}} f = \|\nabla f\| \cos\theta$, the maximum rate occurs when $\cos\theta = 1$ ($\theta = 0$), pointing in the exact direction of ∇f.

11 What is the maximum value of the directional derivative $D_{\mathbf{u}} f$ at a point?

a. $\|\nabla f\|$ ✓

b. $\|\nabla f\|^2$

c. 0

d. 1

When $\mathbf{u}$ points in the direction of ∇f, $D_{\mathbf{u}} f = \nabla f \cdot \frac{\nabla f}{\|\nabla f\|} = \|\nabla f\|$.

12 In which direction is the rate of change of f equal to zero?

a. **In any direction tangent to the level curve (orthogonal to ∇f)** ✓

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

When $\mathbf{u}$ is orthogonal to ∇f, $\cos\theta = 0$, so $D_{\mathbf{u}} f = 0$, which occurs along tangent directions to level curves.

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